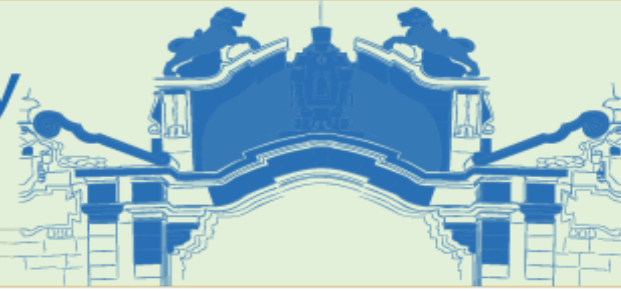




SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !

Addis Ababa University  
አዲስ አበባ ዩኒቨርሲቲ



Addis Ababa Institute of Technology

School of Mechanical and Industrial  
Engineering

Engineering Materials I  
(MEng 2122)

*Chapter - One*

Introduction to Materials Science &  
Engineering

*Contact information;*

*Course Instructor: Ayalsew Dagne*

*Office: 314 C*

*Email: [ayoudag@gmail.com](mailto:ayoudag@gmail.com)*

# Overall Course Objectives

- Understand the basic concepts and language of Materials Science and Engineering.
- Identify major types of materials (metals, ceramics, polymers, composites).
- Understand basic chemistry and structure of materials.
- Understand properties of materials (mechanical, thermal, electrical, chemical, ...).
- Understand different levels of structure in materials (atomic, microscopic, macroscopic).
- Understand imperfections, defects and failure of materials.
- Understand concept and laws of diffusion in materials.
- Understand processing, structure, properties, and performance of materials.
- Understand dislocations and strengthening mechanisms of materials.
- Understand phase diagram.

# Introduction to Materials Science and Engineering

## Basic Contents of the Chapter

- **Historical Background**  
Stone → Bronze → Iron → Advanced materials
- **Why Study Materials Science and Engineering?**  
Processing → Structure → Properties → Performance
- **Classification of Materials**  
Metals, Ceramics, Polymers, Composites,  
Semiconductors
- **Advanced Materials**  
Electronic materials, superconductors, etc.
- **Advanced/Modern Material's Needs, Material of Future**  
Biodegradable materials, Nanomaterials, “Smart” materials

# Chapter \_ Learning Objectives

After completing this chapter, you should be able to:

- Understand and state the different property classifications of materials that determine their applicability.
- State the four components that are involved in the design, production, and utilization of materials, and briefly describe the interrelationships between these components.
- Identify the three criteria that are important in the materials selection process.
- Understand the three primary classifications of solid materials, and then state the distinctive chemical feature of each.
- Note the four types of advanced materials and, for each, its distinctive feature(s).
- Briefly define smart material system.
- Briefly explain the concept of *nanotechnology* as it applies to materials.

# Historical Background of Materials

- Interrelation of our environment with Materials: **transportation, housing, clothing, communication, recreation, food production** and so on in every daily activity.
- The **development** and **advancement** of materials intimately tied to human needs.
- Beginning of the Material Science - people began to make tools from **stone** – Start of the **stone-age** about two million years ago.
- The earliest humans had limited sources of natural materials like; **stone, wood, clay, skins, horn**, etc.
- The **stone-age** ended about 5000 years ago with introduction of **Bronze** in the Far East. **Bronze** is an **alloy**; copper, tin + other elements.
- **Bronze**: can be hammered or cast into a variety of shapes, can be made harder by alloying.

# Historical Perspective

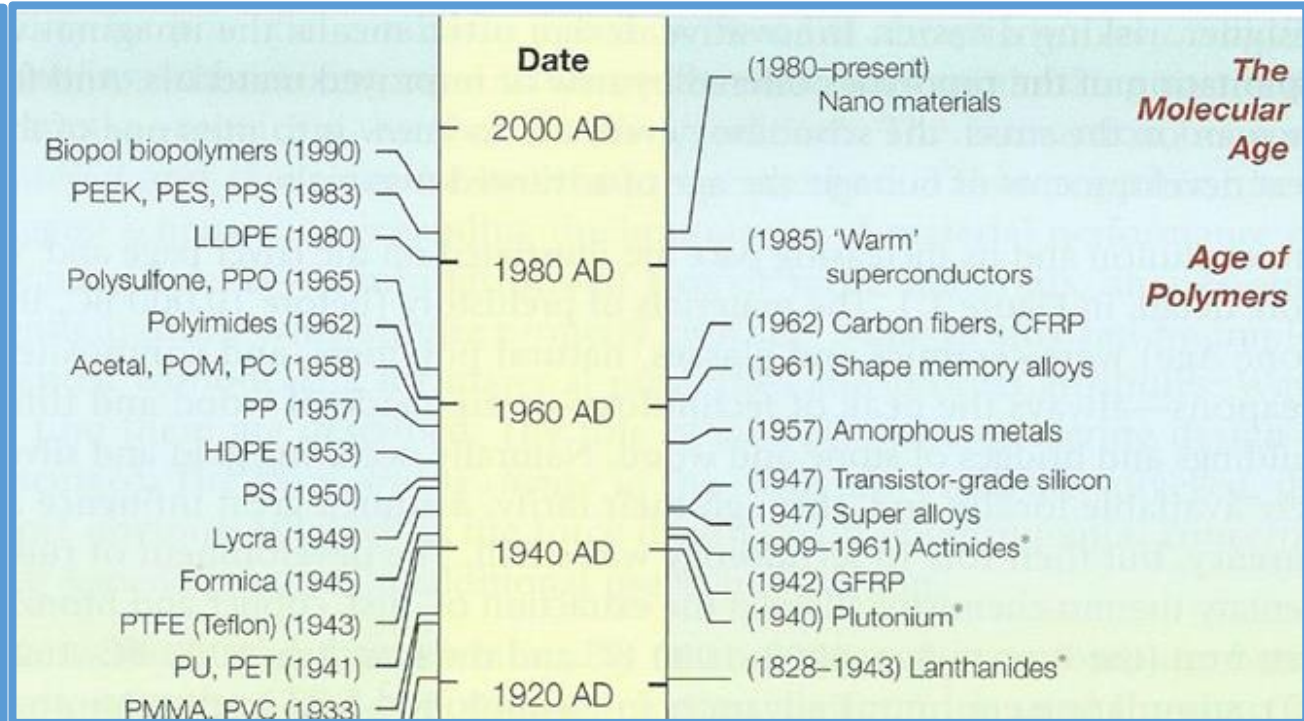
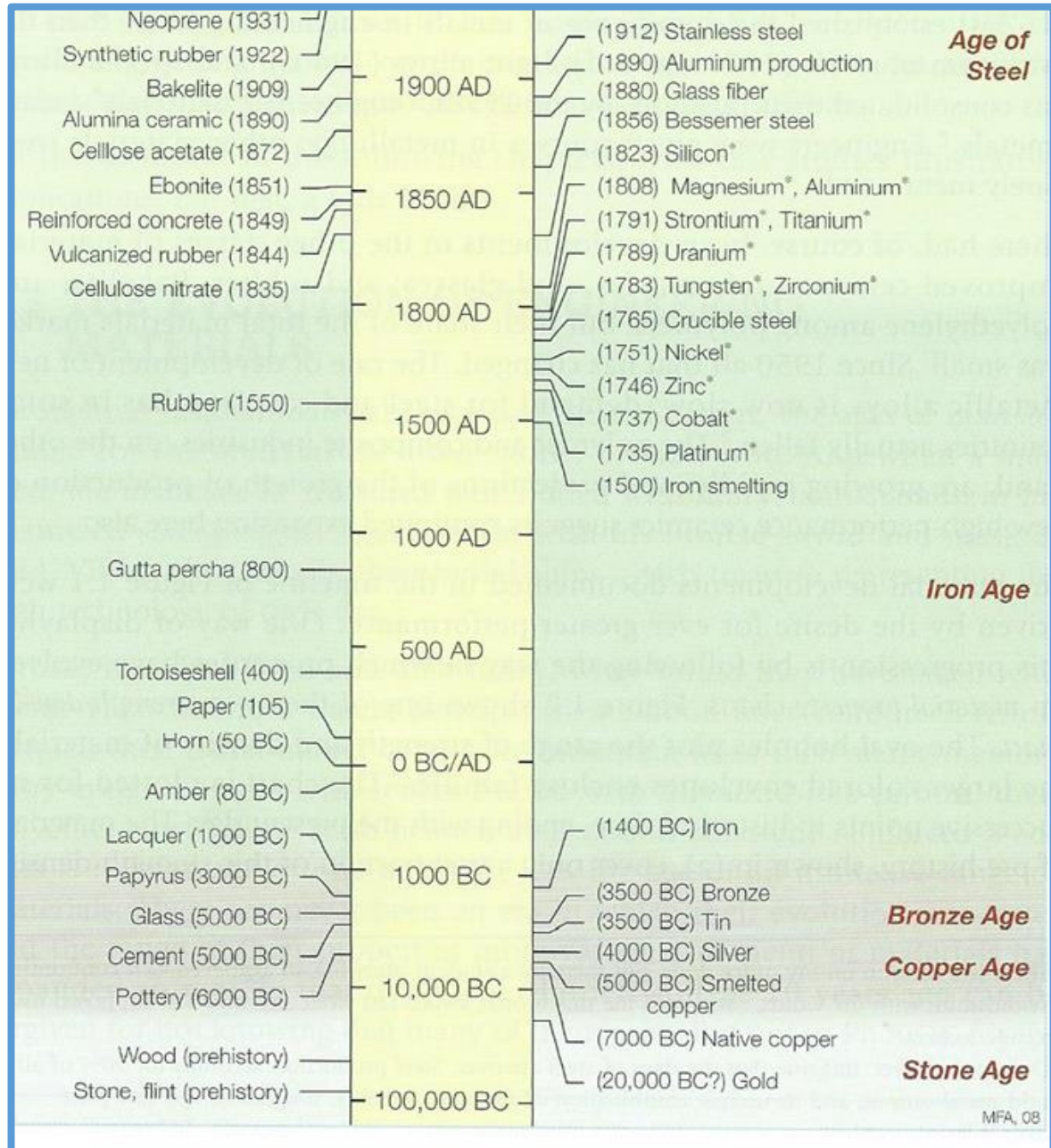
Therefore, this period was called the **bronze-age**.

- The “**Iron-Age**” began about 3000 years ago and continues till today. Use of iron and steel, a stronger and cheaper material changed drastically daily life of a common person. This period is called the **Iron-Age**.
- Age of **advanced materials**: throughout the **iron-age** many new types of materials have been introduced (ceramic, polymers, composites, & semiconductors).

Therefore, the materials advancement looks like;

stone-age → bronze-age → iron-age → advanced/modern-age → future generation

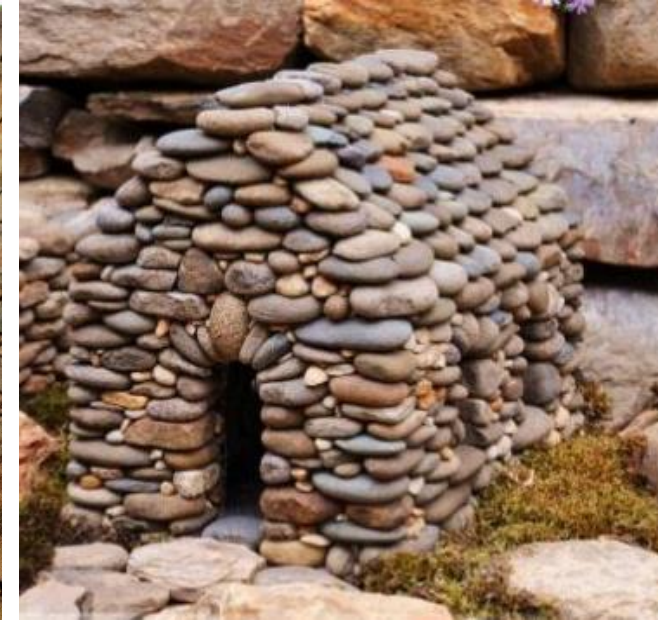
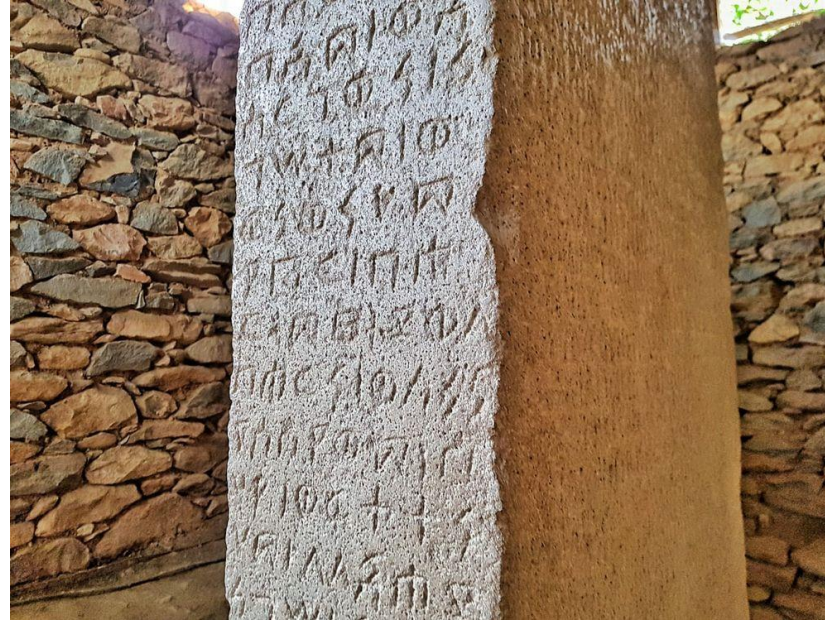
# Historical Perspective



A materials timeline. The scale is nonlinear, with big steps at the bottom, small ones at the top. An asterisk (\*) indicates the date at which an element was first identified. Labels without asterisks note the time at which the material became of practical importance.

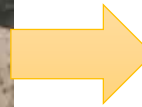
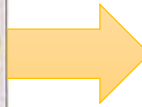
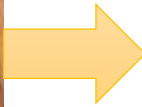
# Historical Perspective

- Historical materials;



# Historical Perspective

- Materials advancement;



# What is Materials Science and Engineering?

## ■ Materials Science:

- Is a discipline of **studying** and **investigating** the relationships that exist between the structures and properties of materials.

## ■ Materials Engineering:

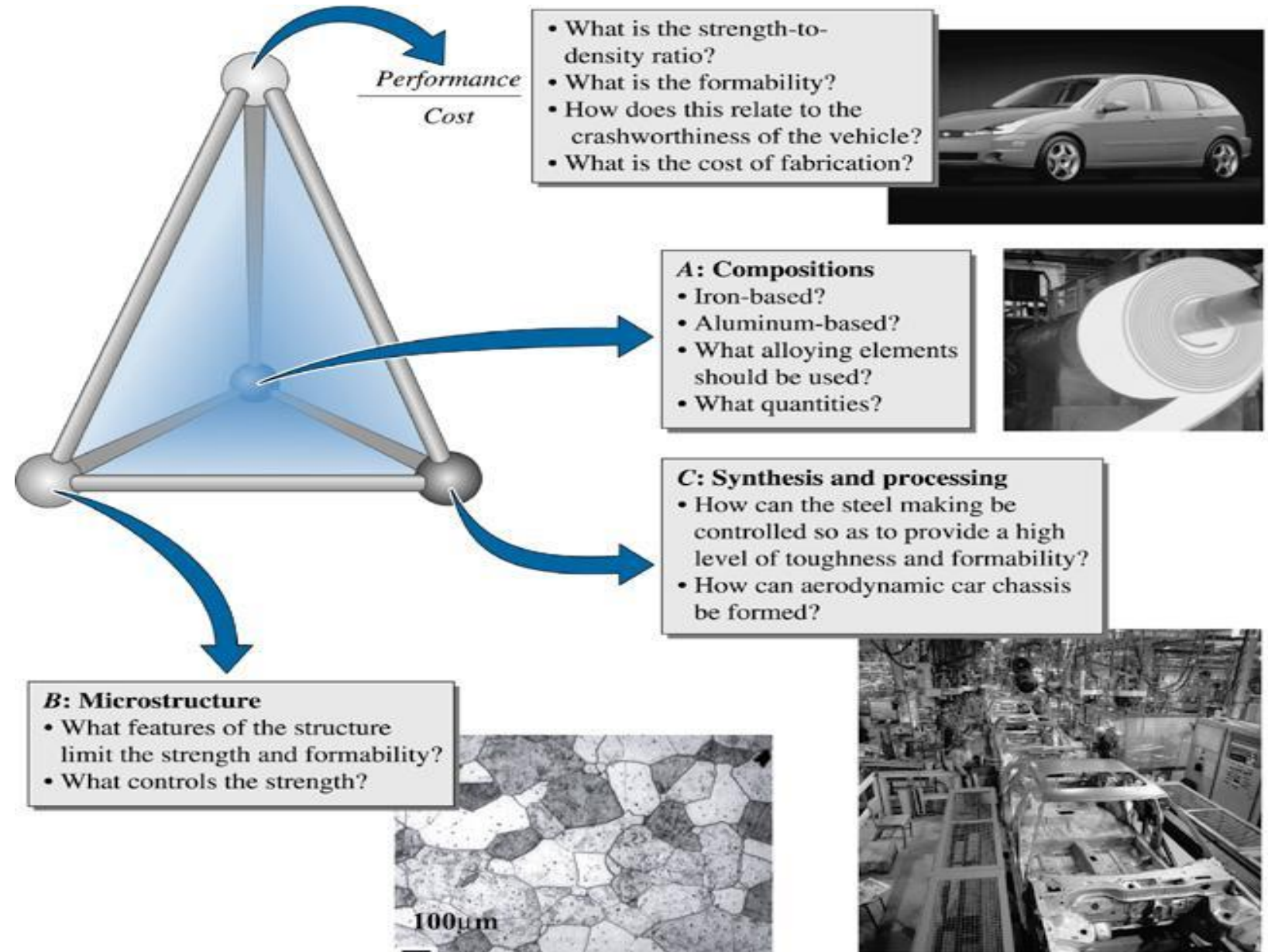
- Is a discipline of **designing** and **processing** the structure of a material to produce a predetermined set of properties based on established structure-property correlation.

♣ **Materials Science and Engineering:** is the discipline of studying the relationships between the structure and properties of materials, and designing to produce.

# Why we Study Materials Science & Engineering?

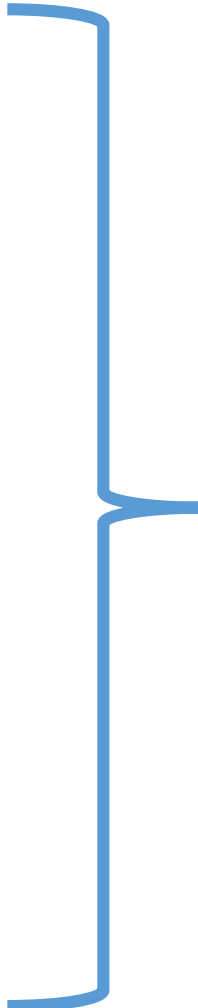
- An illustration to show the interconnection of the basic principles of materials like; microstructures, compositions, synthesis & processing, and their effect on end product performance.

Figure 1.2 Application of the tetrahedron of materials science and engineering to sheet steels for automotive chassis.



# Why we Study Engineering Materials?

- An engineer may be faced to a design problem involving materials, such as the;
  - Transmission gear,
  - Aircraft structure,
  - Car suspension system,
  - Superstructure for a building,
  - Dam and highway structure,
  - Oil refinery component,
  - Pressure vessel & piping structure,
  - Integrated circuit chip,
  - Transformer or other engineering systems.



*To answer and solve such problems it obliged us to study “Engineering Materials”*

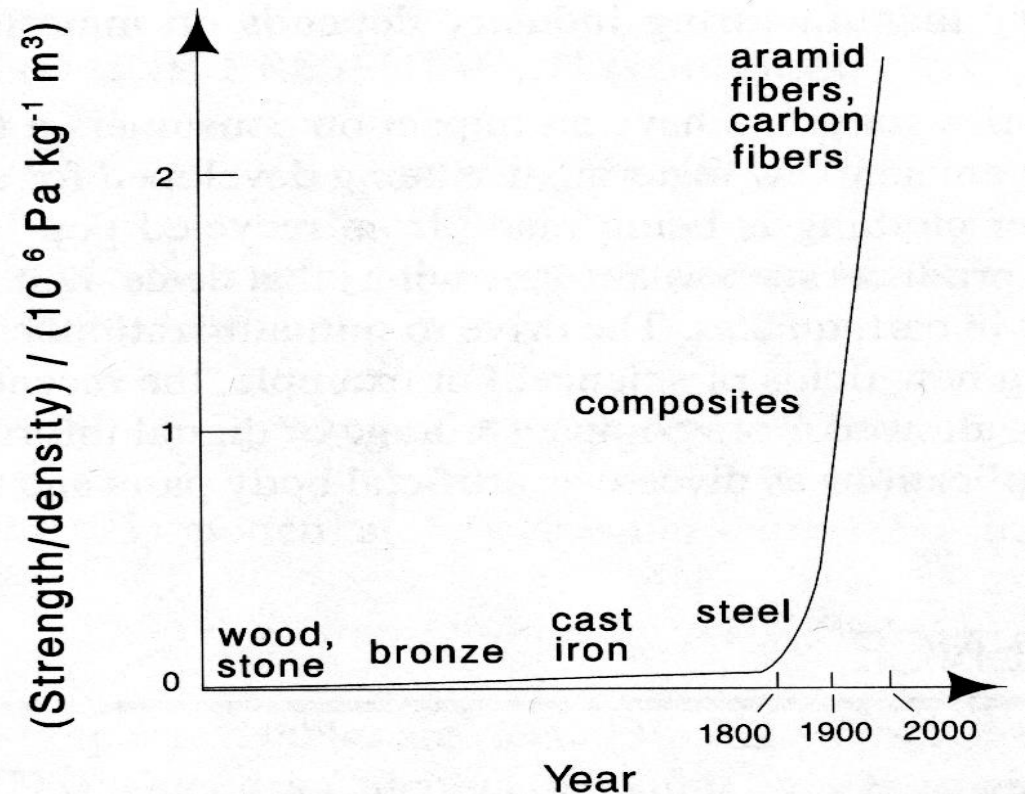
# Why we Study Engineering Materials?

- A **materials problem** is one of selecting the right material from the available materials. The final decision is normally based on several criteria:
  - First, the in-service conditions must be characterized.
  - Second, consideration is any deterioration of material properties that may occur during service operation.
  - Finally, probably the overriding consideration is that of economics.
- Therefore, **an engineer** or **scientist** is with the various **characteristics** and **structure–property** relationships, as well as the **processing techniques** of materials, and providing the more proficient and confident materials choices based on these criteria.

# Why we Study Engineering Materials?

- A better understanding of structure-composition-properties relations has lead to a remarkable progress in properties of materials.

Example is the dramatic progress in the strength to density ratio of materials, that resulted in a wide variety of new products, from dental materials to tennis racquets.



# Why we Study Engineering Materials?

- The basic target of studying engineering materials is/are:
  - Identify type of materials and their properties
  - Identify the most suitable material for a specific application
  - Develop a new material which best suits in required mechanical property
  - Develop a new material which best suits in weight to volume ratio
  - Develop a new material which best suits the specific working environment
  - Develop a new material which best suits in economic point of view
  - Develop a new material which best suits in durability and lifetime
  - Develop a new material which best suits in ergonomic point of view
  - Modify a material which best suits in all human needs
  - Develop and modify a material which best suits in all material properties.

# Properties of Materials

- Materials may be measured in terms of their:

- Physical property

Properties are the way the material responds to the environment and external forces.

- Chemical property

**Mechanical** properties – response to mechanical forces, strength, etc.

- Electrical property

**Chemical** properties – related with response to chemical effects, corrosion resistance, etc.

- Magnetic property

**Electrical** properties - response to electrical conductivity.

**Magnetic** properties – response to magnetic field & poles.

- Thermal property

**Thermal** properties - are related to transmission of heat and heat capacity.

- Optical property

**Optical** properties - include to absorption, transmission and scattering of light.

- Acoustic property

**Acoustic** properties – are related to transmission, absorption and scattering of sound energy.

# Measurement of Materials

- Physical or Mechanical properties measurement:
  - Length measurement – scales for measuring length;

- Angstrom =  $1\text{\AA} = 10^{-10}\text{ m}$
- Nanometer =  $1\text{ nm} = 10^{-9}\text{ m}$
- Micrometer =  $1\mu\text{m} = 10^{-6}\text{ m}$
- Millimeter =  $1\text{mm} = 10^{-3}\text{ m}$

TABLE 1.5 Selected Prefixes Used in the Metric System

| Prefix | Abbreviation | Meaning    | Example                                                      |
|--------|--------------|------------|--------------------------------------------------------------|
| Giga   | G            | $10^9$     | 1 gigameter (Gm) = $1 \times 10^9\text{ m}$                  |
| Mega   | M            | $10^6$     | 1 megameter (Mm) = $1 \times 10^6\text{ m}$                  |
| Kilo   | k            | $10^3$     | 1 kilometer (km) = $1 \times 10^3\text{ m}$                  |
| Deci   | d            | $10^{-1}$  | 1 decimeter (dm) = $0.1\text{ m}$                            |
| Centi  | c            | $10^{-2}$  | 1 centimeter (cm) = $0.01\text{ m}$                          |
| Milli  | m            | $10^{-3}$  | 1 millimeter (mm) = $0.001\text{ m}$                         |
| Micro  | $\mu^a$      | $10^{-6}$  | 1 micrometer ( $\mu\text{m}$ ) = $1 \times 10^{-6}\text{ m}$ |
| Nano   | n            | $10^{-9}$  | 1 nanometer (nm) = $1 \times 10^{-9}\text{ m}$               |
| Pico   | p            | $10^{-12}$ | 1 picometer (pm) = $1 \times 10^{-12}\text{ m}$              |
| Femto  | f            | $10^{-15}$ | 1 femtometer (fm) = $1 \times 10^{-15}\text{ m}$             |

<sup>a</sup>This is the Greek letter mu (pronounced “mew”).

Example; the bonding (interatomic) distance between two atoms is measured in “Angstrom”.

- Interatomic distance  $\sim$  a few  $\text{\AA}$
- A human hair is  $\sim 50\text{ }\mu\text{m}$
- Elongated bumps that make up the data track on CD are  $\sim \mu\text{m}$  and nm level.

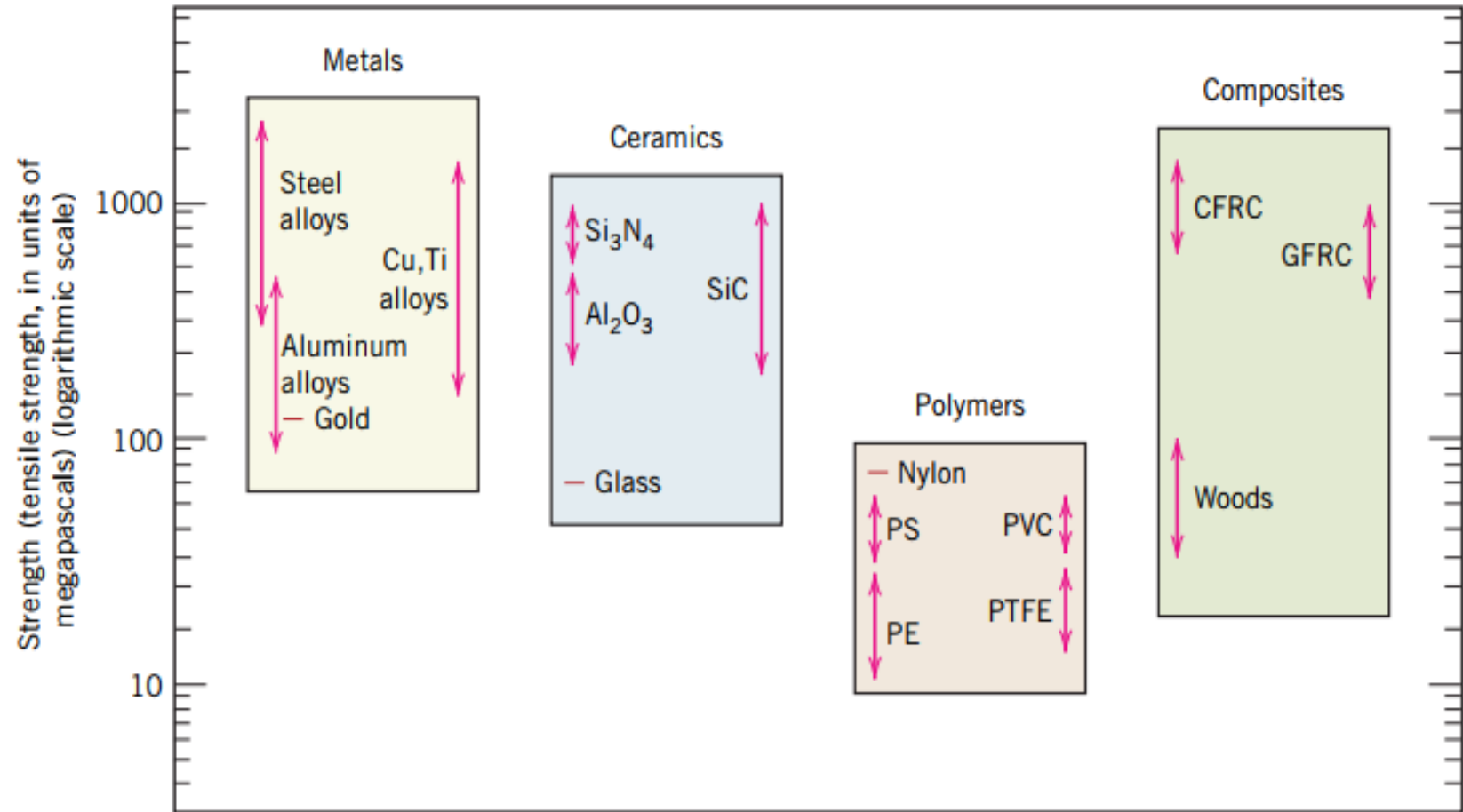
# Classification of Materials

- Based on chemical makeup and atomic structure, solid materials have been conveniently grouped into three basic categories:
  - 1) Metals,
  - 2) Ceramics, and
  - 3) Polymers.
- In addition, there are the composites that are engineered combinations of two or more of the three above materials.
- Another category is advanced materials - those used in high-technology applications, such as semiconductors, biomaterials, smart materials, and nanoengineered materials.

# Classification of Materials

Generally, materials are classified into the four major groups;

- Metals
- Ceramics
- Polymers and
- Composites



**Figure:** Bar chart of room temperature **tensile strength** values for various metals, ceramics, polymers, and composite materials.

# Metals

*Metals* are composed of one or more metallic elements, and often also nonmetallic elements in relatively small amounts.

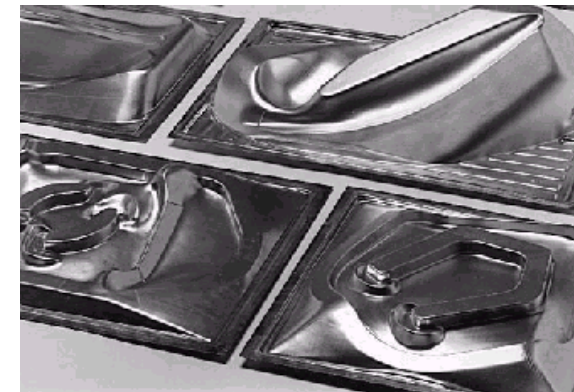
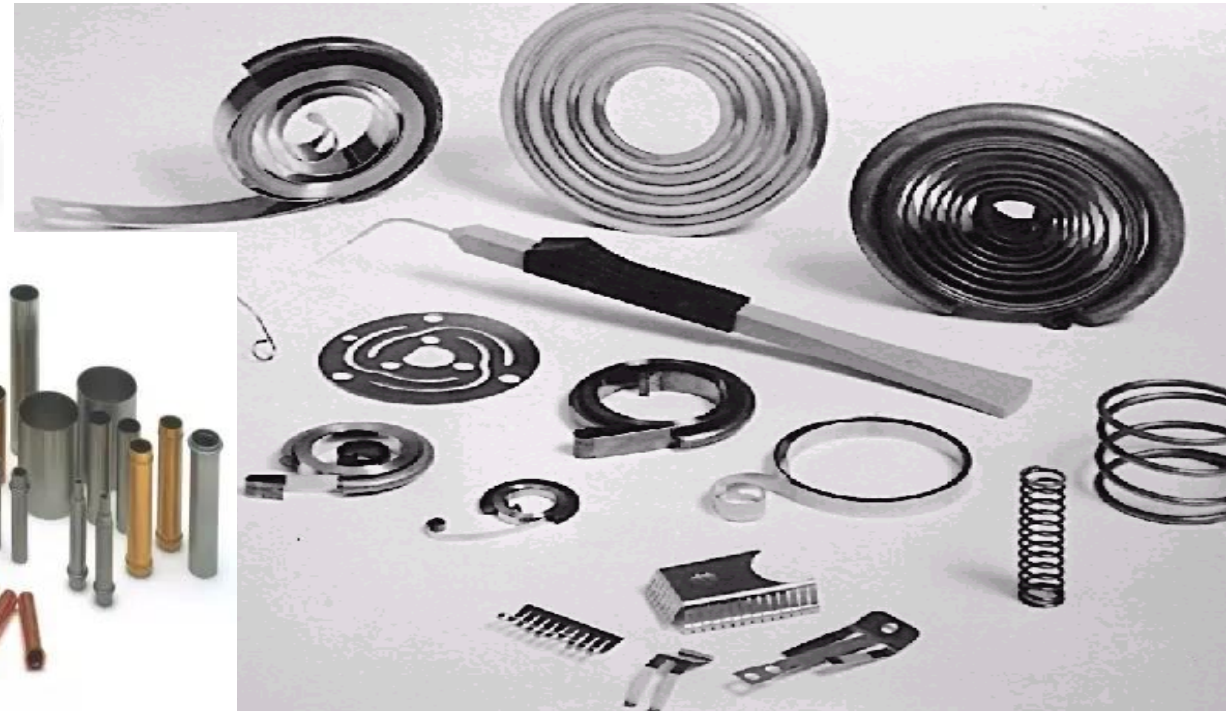
- metallic elements; e.g., iron, aluminum, copper, titanium, gold, nickel
  - nonmetallic elements; e.g., carbon, nitrogen, oxygen
- Atoms in **metals** and their alloys are arranged in a very **orderly manner** and are relatively **dense** in comparison to the **ceramics** and **polymers**.
  - **Metallic materials** have large numbers of **nonlocalized electrons** - that is, these electrons are **not bound** to particular atoms.
  - Many **properties** of metals are directly attributable to these **nonlocalized electrons**.  
For example, **metals** are extremely **good conductors** of electricity and heat, strong, ductile, and shiny if polished.

# Metals

There are different types of metals; in general they can be *ferrous* & *non-ferrous*.

Some examples are:

- Iron
- Cast iron
- Aluminum
- Aluminum alloys
- Steels
- Steel alloys
- Nickel & its alloys
- Titanium, etc.



# Ceramics

**Ceramics** are compounds between **metallic** and **nonmetallic** elements; they are most frequently **oxides**, **nitrides**, and **carbides**.

- common ceramic materials includes;
  - Aluminum oxide (or *alumina*,  $\text{Al}_2\text{O}_3$ ),
  - Silicon dioxide (or *silica*,  $\text{SiO}_2$ ),
  - Silicon carbide ( $\text{SiC}$ ),
  - Silicon nitride ( $\text{Si}_3\text{N}_4$ ),
  - clay minerals (e.g., porcelain),
  - Cement and
  - Glass.

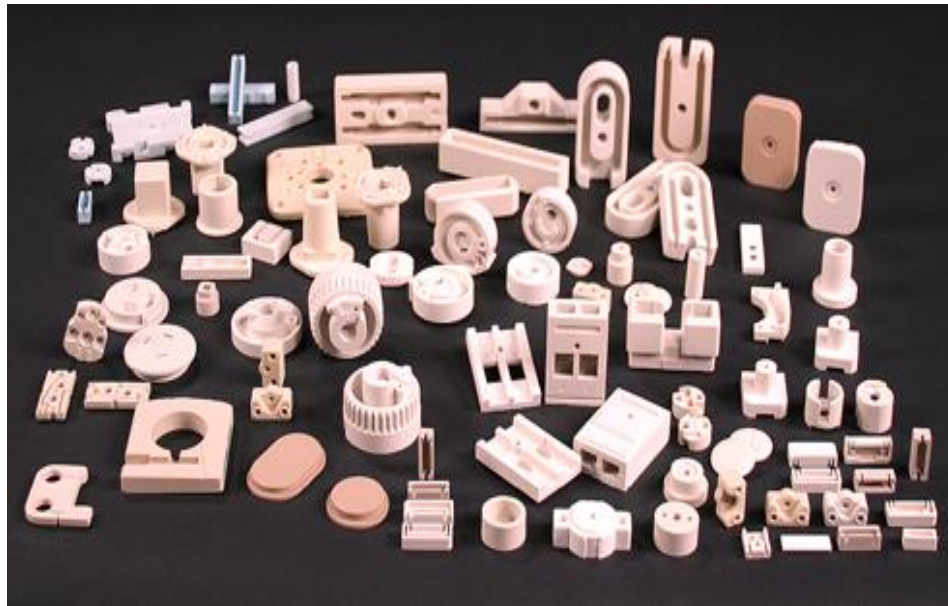


**Figure:** Common objects made of ceramic materials: scissors, a china teacup, a building brick, a floor tile, and a glass vase.

# Ceramics

- **Ceramic materials** are relatively stiff and strong - stiffnesses and strengths are comparable to those of the metals.
- **Ceramics** are high temperature resistant materials.
- **Ceramics** – are extremely brittle and highly susceptible to fracture.

Application of advanced ceramics:



# Polymers

*Polymers* are the familiar plastic and rubber materials.

- Many of them are organic compounds that are chemically based on carbon, hydrogen, and other nonmetallic elements (i.e., O, N, and Si).
- They have very large molecular structures, often chainlike in nature, that often have a backbone of carbon atoms.
- Some common and familiar polymers are;
  - polyethylene (PE),
  - nylon,
  - poly(vinyl chloride) (PVC),
  - polycarbonate (PC),
  - polystyrene (PS), and silicone rubber.



**Figure:** Common objects made of polymeric materials: plastic; spoon, knife, billiard ball, wheel, milk carton, helmet & tires.

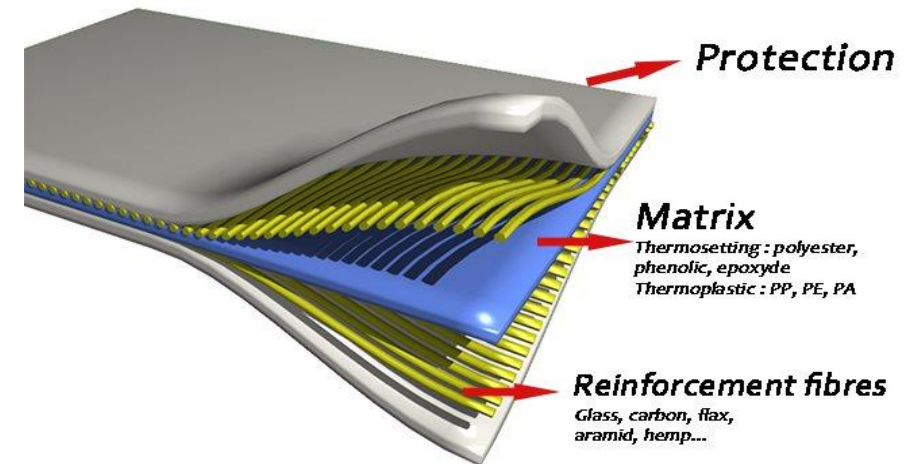
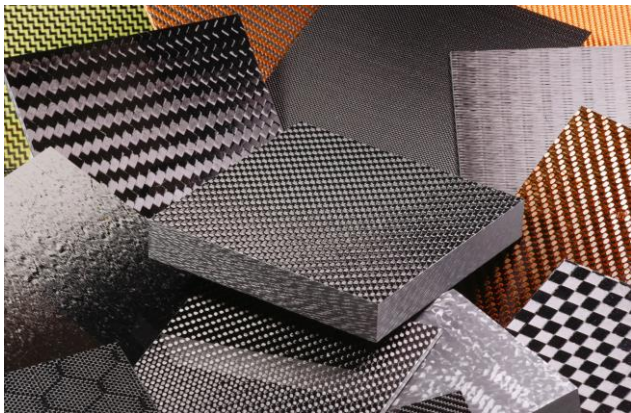
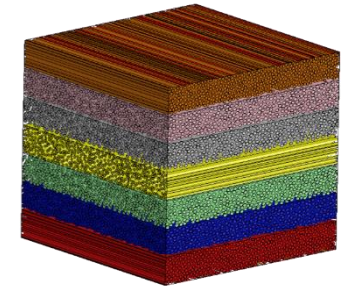
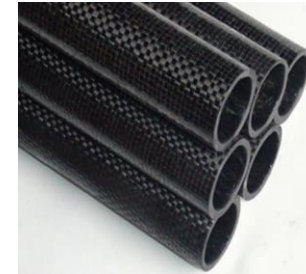
# Composites

**Composites** are composed of two or more individual materials that come from; **metals**, **ceramics**, and **polymers**.

- The **design goal** of a **composite** is to achieve a combination of properties that is not displayed by any single material and also to incorporate the best characteristics of each of the component materials.
- A large number of **composite types** are represented by different combinations of **metals**, **ceramics**, and **polymers**.
- Some common and familiar composites are;
  - **Fiberglass**: is relatively stiff (but also brittle), strong & flexible.
  - **Carbon fiber–reinforced polymers (CFRP)**: are stiffer and stronger than glass fiber–reinforced materials.

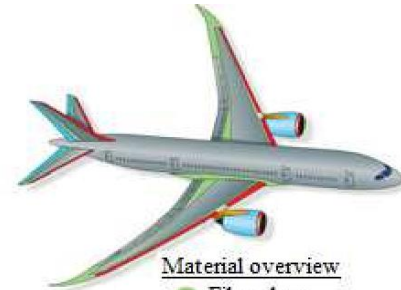
# Composites

- CFRP composites are used in some;
  - Aircraft and aerospace applications,
  - High-tech sporting equipment (e.g., bicycles, golf clubs, tennis rackets, skis/snowboards) and
  - Recently in automobile bumpers.



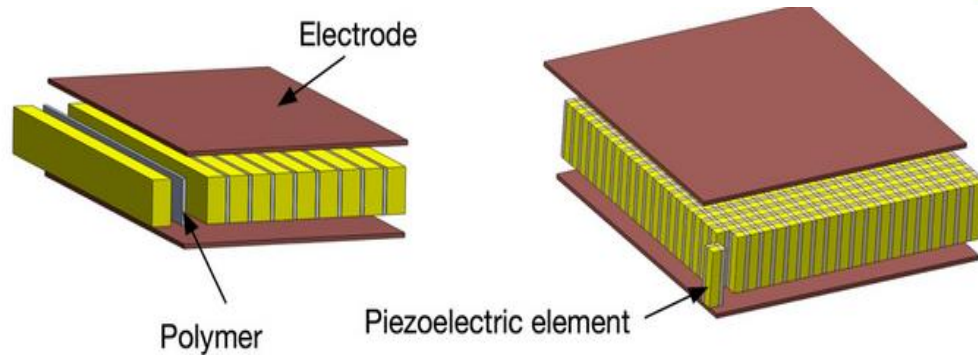
# Composites

## ■ Advanced application of composites

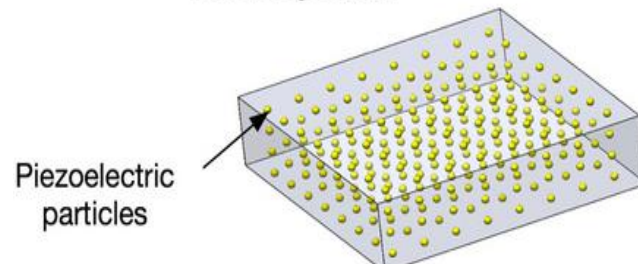


Material overview

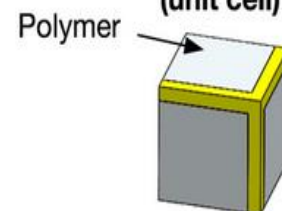
- Fiberglass
- Aluminum
- Carbon laminate composite
- Carbon sandwich composite
- Aluminum/steel/titanium



0-3 composites



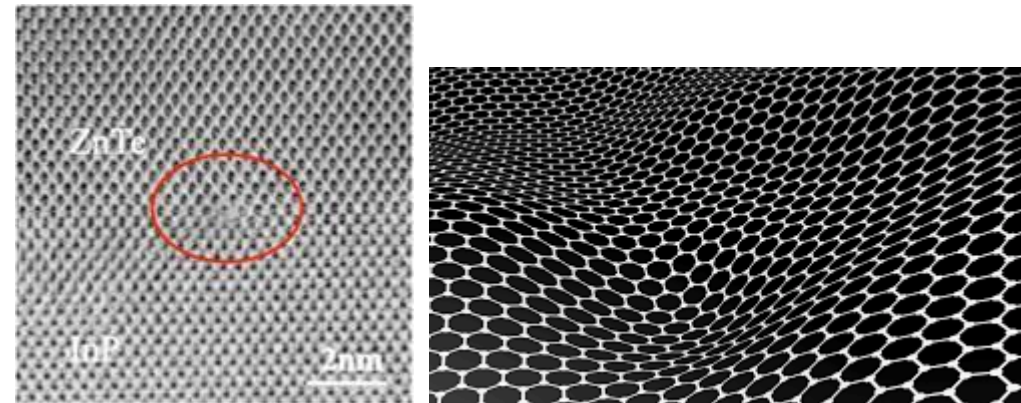
3-3 composites  
(unit cell)



# Advanced Materials

*Advanced Materials* are modern materials utilized in high-technology (or high-tech) applications.

- High technology means a device or product that operates or functions in relatively **intricate** and **sophisticated** principles, including;
  - Electronic equipment (camcorders, CD/DVD players),
  - Computers,
  - Fiber-optic systems,
  - Spacecraft,
  - Aircraft, and military rocketry.
- Advanced materials may be of **all material types** (e.g., metals, ceramics, polymers) and are normally expensive.



# Advanced Materials

- Advanced materials include; semiconductors, biomaterials, smart materials and nanoengineered materials.
- The properties and applications of a number of these advanced materials—for example, materials that are used for;
  - lasers,
  - integrated circuits,
  - magnetic information storage,
  - liquid crystal displays (LCDs), and
  - fiber optics



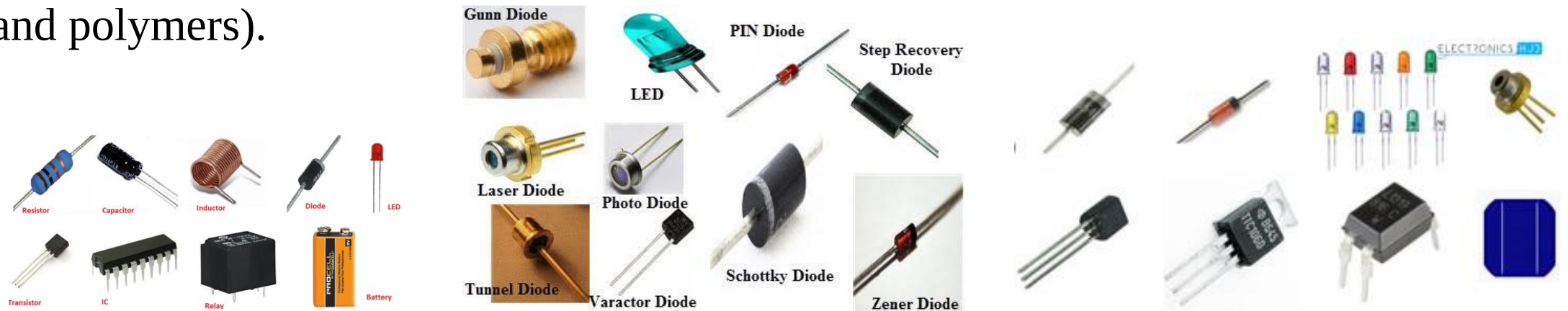
# Advanced Materials

*Semiconductors* are materials formed by **covalent bonding** (electrons are shared between atoms).

- The electrical characteristics of these materials are extremely sensitive to the presence of minute concentrations of impurity atoms.

Examples: Si, Ge, GaAs.

- *Semiconductors* have electrical properties that are intermediate between those of electrical conductors (i.e., metals and metal alloys) and insulators (i.e., ceramics and polymers).



# Advanced Materials

**Biomaterials** are employed in components implanted into the **human body** to replace diseased or damaged body parts.

- These materials must not produce toxic substances and must be compatible with body tissues.
- All of the preceding materials - metals, ceramics, polymers, composites, and semiconductors may be used as biomaterials.

**Smart Materials** *smart (or intelligent) materials* are a group of new and state-of-the-art materials now being developed that will have a significant influence on many of our technologies. Smart material includes; **sensors** and **actuators**.

**Nanomaterials** are class of modern materials that has fascinating properties and tremendous technological advancement. Basic; metals, ceramics, polymers, or composites.

# Advanced Materials

Some of the applications of nanomaterials are the following:

- Catalytic converters for automobiles.
- Nanocarbons — fullerenes, carbon nanotubes, and graphene.
- Particles of carbon black as reinforcement for automobile tires.
- Nanocomposites.
- Magnetic nanosize grains that are used for hard disk drives.
- Magnetic particles that store data on magnetic tapes.

# Advanced Materials

**Modern Materials Need's** are due to:

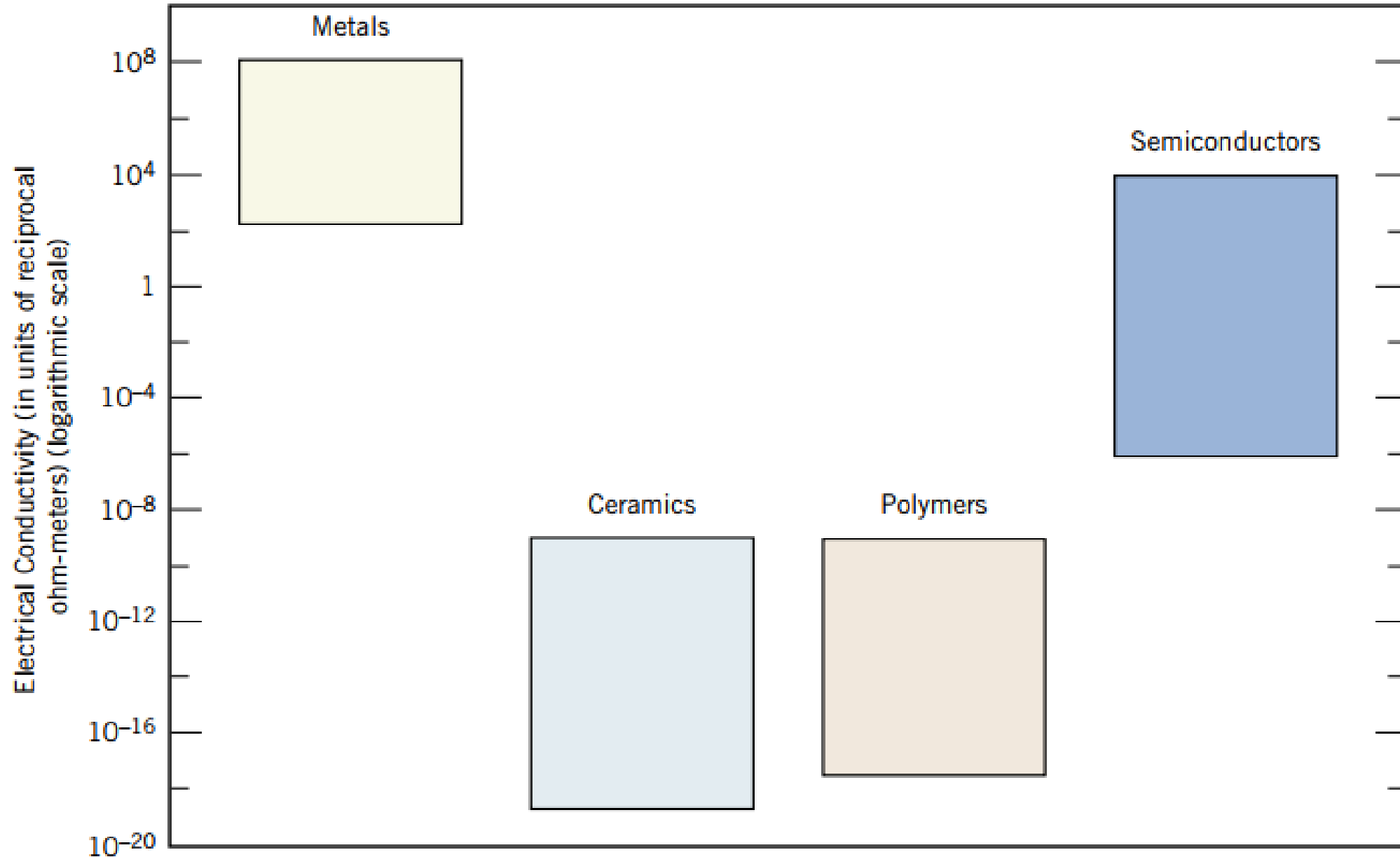
- The need of renewable energy resources
- Nuclear energy generation
- Quality improvement
- Cost improvement
- Nano-size improvement and other reasons.

**Processing – Structure – Properties → Correlations**

- Processing – Structure – Properties → Correlates the processing of materials with their microstructural design and properties for their efficient application.

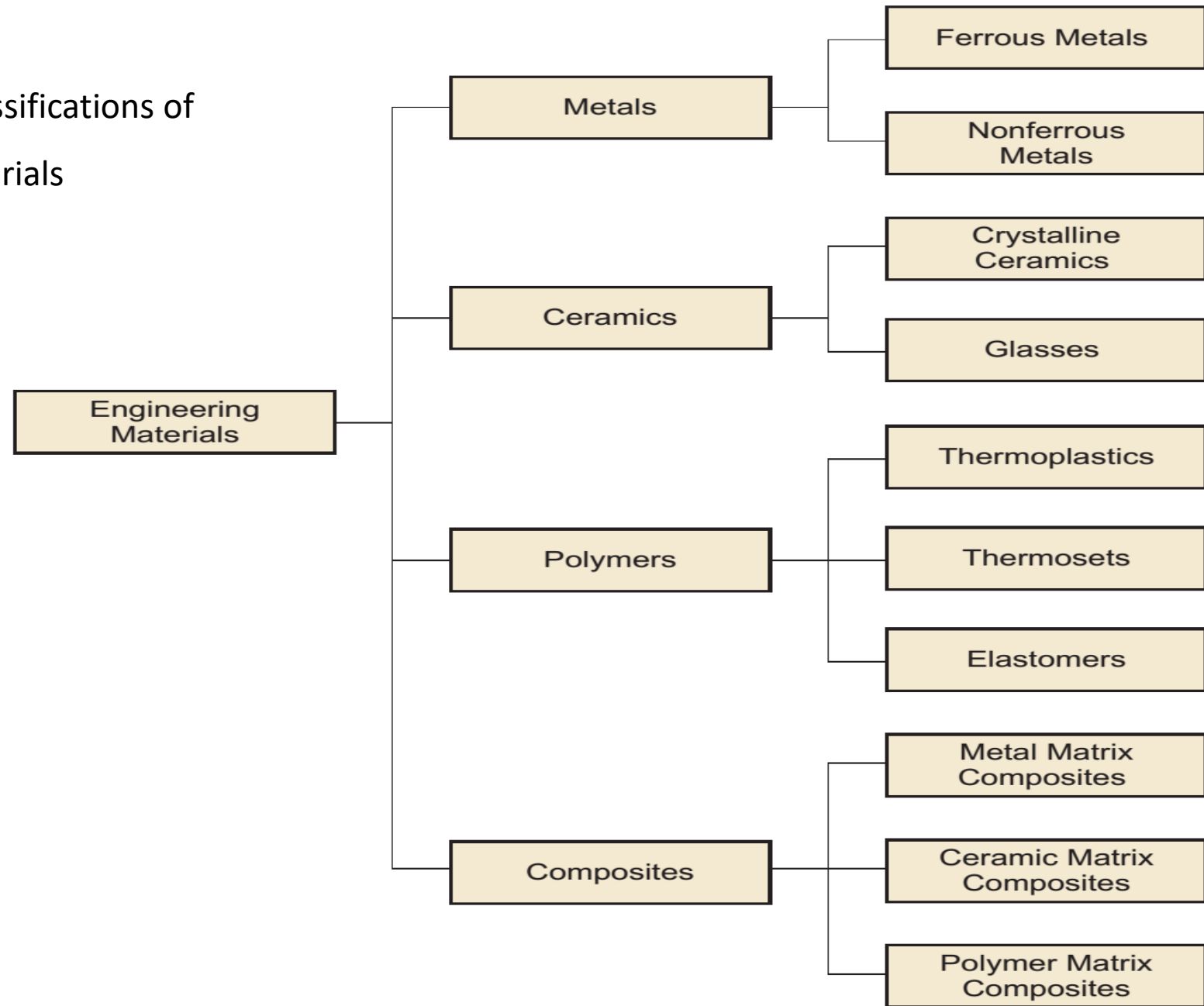
# Summary

**Figure:** Bar chart of room-temperature electrical conductivity ranges for metals, ceramics, polymers, and semiconducting materials.



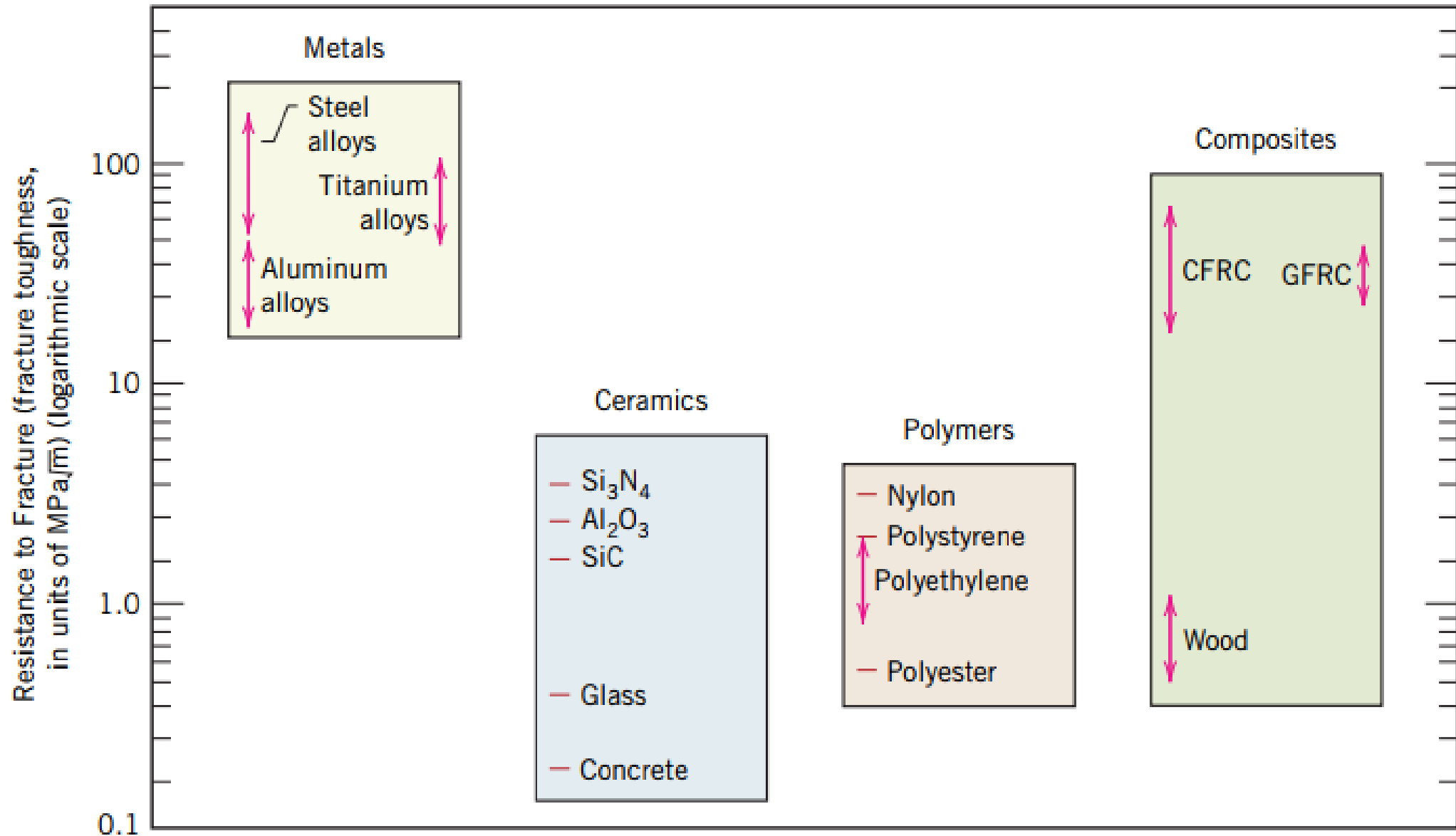
# Summary

Figure: Major classifications of engineering materials



# Summary

**Figure:** Bar chart of room-temperature resistance to fracture (i.e., fracture toughness) for various metals, ceramics, polymers, and composite materials.



*End of the Chapter!*

*Thank you!*